

Food Classification Using Deep Learning

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1. Introduction

Automated food recognition plays an important role in applications such as dietary tracking, smart kitchens, healthcare monitoring, and automated service systems. However, fine-grained food classification is challenging because many dishes share similar textures, colors, and ingredients; visual differences are often subtle and highly sensitive to lighting, viewpoint, and presentation.

This project addresses this challenge by developing a deep learning-based food classification system using the Food-101 dataset. We evaluate four models: (1) a baseline linear-probing model with a frozen ResNet-18 backbone, (2) a baseline model that additionally fine-tunes the final convolutional layer, (3) an improved model that incorporates Squeeze-and-Excitation attention, and (4) the same improved model with stronger data augmentation and regularization. Along the way, we make various modifications to understand the effectiveness of each technique and its role in building a better model. Each model's performance is compared quantitatively and qualitatively.

2. Problem Motivation

Food recognition is becoming increasingly important in a variety of modern contexts such as calorie tracking apps, automated checkout systems, food delivery robots, and nutrition analysis tools. Manually logging what we eat is tedious and error-prone, both issues that automation helps to alleviate. In the future, autonomous service agents will likely need strong perception models that can recognize meals on real plates in real-world settings. Hence, improving food recognition not only benefits current user-focused applications but also lays the groundwork for more advanced AI-driven service systems.

3. Dataset Description

The Food-101 dataset from Bossard et al. (2014) contains 101 food categories with 1,000 images per class, split into approximately 70,000 training images and 30,000 test images. Training labels contain noise, while the test set is manually verified. Images vary widely in resolution, lighting, orientation, and background, making the dataset highly challenging.

All images were normalized using ImageNet statistics. Later as an improvement, data augmentation was applied to the test batch, including random cropping, horizontal flipping, rotation, and color jitter.

4. Methodology

4.1 Baseline Model

Our baseline model is inspired by the methodology presented in the DeepFood paper by Liu et al. (2016), which uses an Inception-based CNN (GoogLeNet) trained on non-augmented Food-101 images. We decided to instead use ResNet-18 as our baseline model to preserve the core idea of fine-tuning a pretrained convolutional network. Specifically, when fine-tuning, we freeze earlier

weights and retrain the classification head to establish performance without additional architectural and methodological innovation.

Training was conducted using Adam optimization and cross-entropy loss. We tracked validation performance and ensured consistent preprocessing with the original DeepFood approach.

4.1.1 Fully Connected Classifier

For our first baseline to compare against, we froze all layers except for a final fully connected layer. We did this in order to understand the existing capabilities of the model and the feature extractors learned by the ImageNet pretraining pass. We could later then compare this performance to the performance of improved models and training pipelines to see how the model improved.

4.1.2 Fine-Tuned Classifier

To evaluate whether deeper feature adaptation beyond the ImageNet features alone could improve performance, we ran a second baseline variant in which the final convolutional layer (layer 4) was unfrozen. This modification increases the model's capacity to learn mid-level and high-level structures specific to food imagery while still leveraging the stable low-level filters learned from ImageNet.

This baseline variant allows us to isolate the impact of adapting additional layers to food-specific textures, colors, and ingredient compositions. This provides a controlled comparison point for assessing later architectural additions, such as Squeeze-and-Excitation attention and a more rigorous training pipeline.

4.2 Improved Model

To address the baseline model's underfitting and lack of food-specific feature adaptation, we developed an enhanced architecture that incorporates attention, expanded augmentation, and stronger regularization.

4.2.1 Squeeze-and-Excitation Attention

The first step towards improvement we made was to integrate a Squeeze-and-Excitation (SE) attention module after each residual block. SE is a light-weight attention mechanism designed to improve a convolutional network's ability to focus on informative feature channels (Hu et al., 2018). Unlike convolutional layers, which learn spatially sensitive filters such as edges and textures, SE learns an importance weight for each channel that is applied uniformly across the entire spatial field. This allows the network to adaptively boost or suppress whole channels based on their relevance to the current input.

In order to properly train each SE module, we also fine-tune the entire network so that the SE layers can adjust the incoming feature representations rather than operating on fixed, pretrained activations. This ensures that the channel-wise recalibration learned by SE is aligned with the updated convolutional filters and can

meaningfully influence the model’s high-level representation of food images.

We expect this approach to be particularly effective for food classification because many Food-101 classes are distinguished by global characteristics, such as dominant color, surface texture, or ingredient patterns, rather than a fixed spatial arrangement of features. By recalibrating channels that encode these food-specific cues, SE provides a flexible mechanism for improving high-level discriminative power while keeping computational cost low.

4.2.2 Regularization

The second improved model introduced stronger data augmentation consisting of random cropping, horizontal flipping, rotation, and color jitter. This increases sample diversity and reduces the model’s reliance on narrow or overly specific visual patterns, thereby improving generalization to new images. Strong augmentation helps learn invariance that are essential for a robust classification model.

Additionally, other regularization and optimization strategies were used to further mitigate overfitting. A dropout layer ($p=0.5$) was added prior to the classification head to encourage more robust representations and reduce overfitting. In addition, we adopted a cosine-annealing learning-rate scheduler to promote smoother convergence. Cosine annealing gradually reduces the learning rate following a cosine decay curve, allowing the optimizer to take larger exploratory steps early in training while settling into progressively smaller refinements later on. This schedule often leads to more stable optimization dynamics and helps the model avoid premature convergence to suboptimal minima.

5. Experimental Setup

Experiments were conducted using PyTorch with CUDA acceleration. Experiments used identical train/test splits and consistent preprocessing pipelines to enable fair comparison. Evaluation focused on top-1 accuracy, loss curves, class-level confusion patterns, and qualitative observations from misclassified samples.

6. Results

6.1 Baseline Model Performance

6.1.1 Fully Connected Classifier Performance

Figure 1 shows that the baseline model exhibited smooth but slow convergence. Training accuracy gradually

increased to $\sim 50\%$, while test accuracy reached a maximum of 55.8%. Loss curves indicate that test loss consistently remained below training loss, suggesting that the frozen backbone restricted the model’s capacity and produced underfitting. This behavior is also consistent with the structure of the Food-101 dataset. The training split contains noisier, less curated images, whereas the test set is manually cleaned. As a result, achieving higher accuracy on the test set compared to the training set in an underfitting model is expected and does not indicate overfitting in this particular context.

The baseline confusion matrix reveals widespread low-grade confusion across several food categories, particularly those with similar textures or color palettes. Meat steaks, sandwiches, soups, and desserts frequently exhibited mutual misclassifications.

Overall, the baseline model produced stable but limited performance, struggling to internalize fine-grained distinctions due to the lack of trainable depth and the absence of food-specific feature adaptation.

6.1.2 Fine-Tuned Classifier

Figure 2 shows that the training accuracy for the fine-tuned model increased rapidly from about 30% to nearly 100% by epoch 20, demonstrating effective specialization of high-level features. Test accuracy improves substantially over the fully frozen baseline, reaching a peak of 74.5% within the first 10–12 epochs before plateauing. Training loss steadily approaches zero, while test loss begins to rise after about epoch 10, indicating the onset of overfitting. The confusion matrix shows improved class separation, though visually similar food categories (e.g., pork chops and steaks, chocolate cake and tiramisu) remain challenging.

This confirms that the pre-trained mid-level to high-level features are not optimally aligned with our task, and that even a small amount of fine-tuning amounts to approximately 20% in test accuracy improvement. On the other hand, the success (and subsequent overfitting) clearly demonstrates that the low-level features learned from ImageNet are transferable to our image classification task effectively. However, the plateau in test accuracy and the growing gap between training and test loss indicate that the model is still not emphasizing the most informative aspects of the data. In many Food-101 categories, cues are carried not by spatial structure but by global channel characteristics. Because the fine-tuned

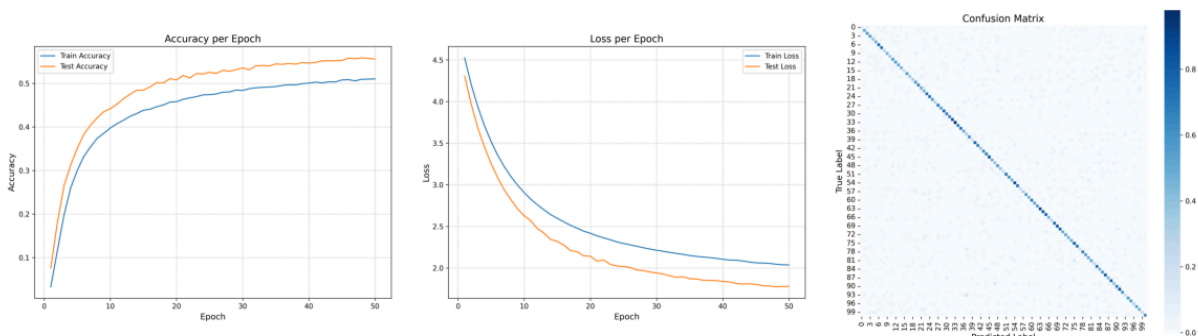


Figure 1. The baseline fully-connected classifier’s performance.

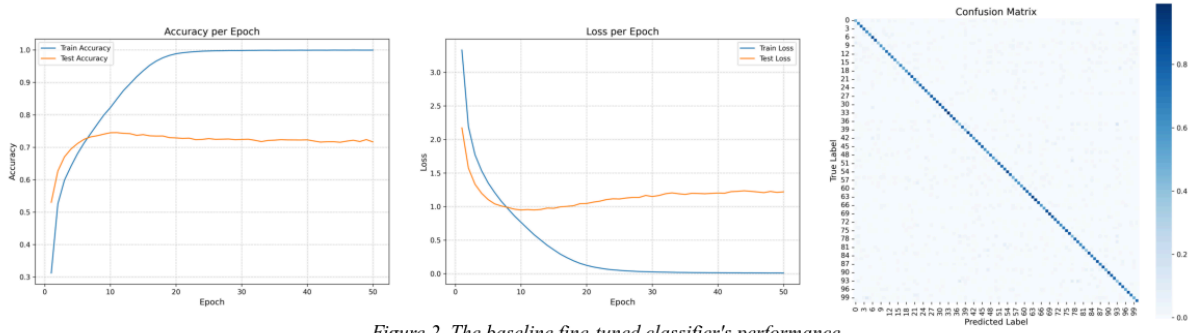


Figure 2. The baseline fine-tuned classifier's performance.

model treats all feature channels equally, it may overfit to spurious or overly specific channels instead of highlighting those that best distinguish food categories.

Overall, this suggests that improving channel-wise feature selection may help generalization. These observations motivate the integration of SE, which may adaptively re-weight channels based on their relevance to each input.

6.2 Improved Model Performance

6.2.1 Squeeze-and-Excitation Attention

To evaluate whether channel-wise recalibration could improve the model's ability to emphasize informative cues, we introduced an SE module after each residual block, consistent with the methodology of the original SE networks paper (Hu et al., 2018). With this modification, the SE-augmented model seemingly performed nearly identically to the fine-tuned baseline in terms of training behavior. Training accuracy, test accuracy, and loss curves followed almost the same trajectory, and the confusion matrix did not exhibit a noticeably stronger diagonal.

However, we did observe a measurable performance improvement. Looking at Figure 3, the peak test accuracy increased from 74.5% to 76.5%, and the training process exhibited slightly more stable loss behavior. The confusion matrix showed modest reductions in confusion among visually similar classes, though not uniformly across all categories. The onset of overfitting was also apparent after 10-15 epochs.

These results are consistent with the original SE paper, which reports that SE blocks typically yield small but reliable gains on the order of 3-5% on ResNet architectures. When added to lightweight architectures such as ResNet-18, SE does not fundamentally change the spatial processing of the network; instead, it improves the utilization of existing features through channel-wise reweighting. Accordingly, the improvements we observe are incremental rather than dramatic, but still meaningful in increasing overall classification accuracy.

6.2.2 Data Augmentation & Regularization

The improved model and training pipeline exhibit both stronger generalization compared to the fine-tuned baseline, illustrated in Figure 4.

Training accuracy no longer rises toward 100%; instead, it stabilizes around 70–75%, while test accuracy reaches a high of 81.2%. This behavior indicates reduced overfitting: the model is prevented from memorizing training examples and instead learns features that transfer more reliably to unseen data.

The confusion matrix shows structural improvements, but the biggest indicator of improvement is the ~5% increase in test accuracy. Additionally, the model exhibits far greater training stability, with reduced divergence between training and test loss. As before, most misclassifications occur between extremely similar categories or images captured under challenging lighting or viewpoints.

These results highlight the impact of the augmented training pipeline on generalization. The augmentation increases sample diversity and prevents the network from relying on overly specific visual cues. Dropout and cosine learning-rate scheduling help regulate capacity and avoid premature overfitting, an improvement illustrated in Figures 2 and 3, where a characteristic drop in test accuracy is seen after early epochs. On the other hand, the model with improved regularization maintains consistent performance throughout training. Together, these additions produce a model that not only achieves higher accuracy but also learns more robust, transferable representations, ultimately outperforming all the previous models.

7. Discussion

This study highlights the strong sensitivity of fine-grained food classification to architectural design and training strategy. The progression from the classifier model to the fine-tuned model, and finally to the fully improved model, reveals that no single modification is sufficient for strong performance on Food-101, but rather the compound

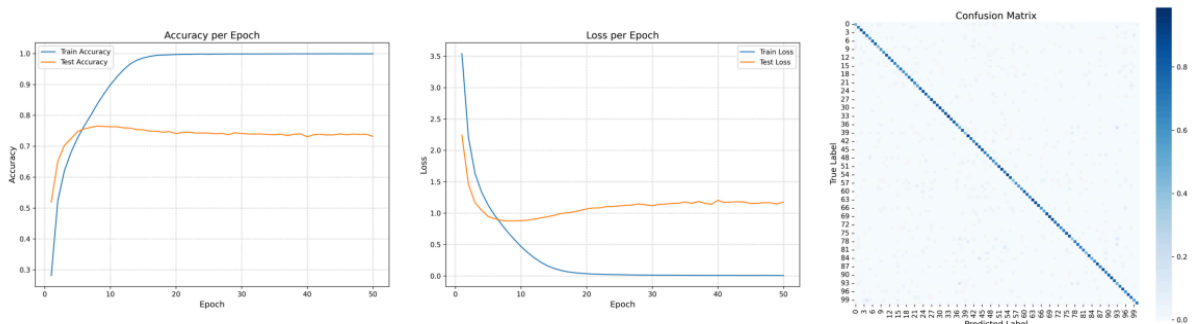


Figure 3. The improved squeeze-and-excitation attention model's performance.

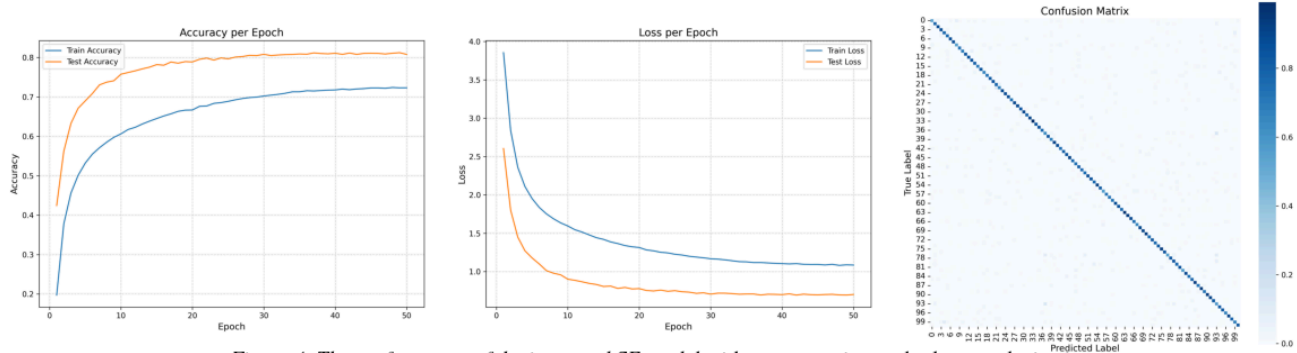


Figure 4. The performance of the improved SE model with augmentation and other regularization.

effects across representation learning, attention, and regularization.

The classification baseline model, while stable and computationally efficient, underfits the Food-101 dataset, exhibiting limited feature adaptation and high interclass confusion. The fine-tuned baseline model clarified the primary bottleneck. Unfreezing even a single deep block increased top-1 accuracy by nearly 20%, showing that food recognition requires adaptable mid-level and high-level features that ImageNet does not provide out of the box. At the same time, the rapid rise of training accuracy toward 100% and divergence from test performance exposed severe overfitting, indicating that deeper adaptation alone is not enough.

Integrating SE attention provided noticeable but modest improvements, consistent with prior work. SE increased peak accuracy from 74.5% to 76.5%, stabilized training, and slightly improved separability for visually similar classes. This aligns with the original SE paper’s finding that channel-wise recalibration yields reliable but incremental gains in lightweight architectures such as ResNet-18. Because SE enhances feature utilization rather than altering spatial processing, its benefits remain constrained by the underlying representational capacity of the backbone.

Augmentation and regularization also yielded significant qualitative improvement. Stronger augmentation diversified the training distribution, reducing reliance on narrow visual patterns and sharply limiting overfitting. Dropout and cosine-annealing learning-rate scheduling further stabilized optimization, producing a model that did not achieve perfect training accuracy but generalized more effectively. These modifications improved robustness and consistency even when accuracy gains were modest.

Together, the results illustrate that fine-grained food classification is not driven solely by architectural complexity but by an interplay of representation adaptation and generalization-oriented training strategies. The remaining misclassifications often involve dishes that are visually similar or rely on ingredient-level cues not readily visible, suggesting limitations inherent to static down-sized imagery.

Future work may explore more transformer-based vision architectures, such as spatial attention instead of traditional convolutional networks (Zhu et al., 2019). Food images often have irrelevant background information that

spatial attention mechanisms can suppress and then focus on the actual informative cues.

Additionally, multimodal learning offers another promising direction. Incorporating textual information such as ingredient lists, recipe metadata, or dish descriptions could provide complementary semantic cues, helping distinguish classes that are visually similar but semantically distinct. Recent work in joint vision-language learning demonstrates the power of combining visual and textual embeddings for stronger classification results (Sun et al., 2023).

Finally, deployment of the model on edge devices may also call for techniques such as model pruning or quantization.

8. Division of Work

Riya Kumar: Improved model training pipeline, initial SE integration, data augmentation, and experimentation logging; led the writing of the final report.

Kyle Vo: Implemented epoch training and test graphs and proposed original ResNet model; improved baseline model; developed the presentation.

Joshua Chasnov: Implemented and revised the baseline and improved architectures; drafted technical report sections; generated all project figures.

Final writing, editing, and interpretation of the code, report, and presentation were completed collaboratively.

9. Conclusion

This project demonstrates that effective food classification accuracy on the Food-101 dataset requires more than simply applying an off-the-shelf ImageNet pre-trained model. Our experiments show substantial improvement from the interaction of several key components: deeper fine-tuning, channel-level mechanisms, enriched data augmentation, and well-chosen regularization strategies. The improved model outperforms the baseline significantly and reduces confusion across numerous visually similar food categories. These findings underscore the importance of adapting network architecture and training strategies to the demands of fine-grained visual tasks.

10. References

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